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Mitigating Phase-Redistribution Effects in Pressure Transient Analysis Using Physics-Guided Neural Networks

M. Corrales, S. Asnin, and M. Ravasi, Physical Sciences and Engineering Division, KAUST, Saudi Arabia; Q. Sahu, H. Qanadeely, M. Alodah, and R. Phillips, EXPEC ARC, Saudi Aramco, Saudi Arabia; H. Hoteit, Physical Sciences and Engineering Division, KAUST, Saudi Arabia

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Summary

This study introduces a new data-driven approach to mitigate the effects of complex wellbore phase-redistribution on buildup tests in vertical wells with high water-cut. Phase-redistribution occurs due to gravity-driven fluid segregation, and alters the distribution of oil and water in the wellbore and leads to unreliable bottom-hole pressure (BHP) measurements at the downhole gauge; thus, Pressure Transient Analysis (PTA) tests become unreliable or can be misinterpreted. First, we develop a coupled reservoir-wellbore black-oil fluid flow model to simulate the effects of phase-redistribution under various operational conditions (including water cut, gauge location relative to the sandface, fluid properties, and reservoir parameters). From these simulations, we assemble a labeled training set pairing distorted pressure curves (affected by phase-redistribution) with the corresponding clean signals at the sandface. Next, we train a 1D U-Net convolutional neural network on the paired data. The network receives as input the distorted pressure signal and the Bourdet derivative, along with a mask highlighting intervals where the effect of phase-redistribution is present. Once trained, the network is tested on synthetic datasets. In the cases evaluated, it successfully restores the true pressure response, although results depend on careful selection of the mask to isolate the distorted zones. By removing phase-redistribution effects upfront, corrected pressure curves can be matched directly to analytical solutions of the diffusivity equation (DE) without additional wellbore storage adjustments. The novelty of the proposed method lies in its ability to remove phase-redistribution effects using a neural network trained on high-fidelity simulations, rather than relying on analytical solutions to approximate changes in wellbore storage for pressure signal matching. This data-driven approach opens new avenues to incorporate additional physical phenomena and extend its application to inclined and horizontal wells.

Introduction

The concept of wellbore storage in PTA was formally introduced by [Van Everdingen & Hurst \(1949\)](#) and [Van Everdingen \(1953\)](#), and provides a foundation for understanding various wellbore phenomena, such as fluid compressibility, afterflow, phase segregation, and redistribution. [Stegemeier & Matthews \(1958\)](#) provided

the earliest field evidence of "hump" buildups caused by phase-redistribution and showed that bottom-hole pressure may exceed the formation pressure and complicate traditional Horner analysis (Horner, 1951). Around the same time, Pitzer et al. (1959) demonstrated that phase-redistribution can significantly distort the surface shut-in buildup curves, which prompted the development of collar lock pressure gauges to enable a more effective bottom-hole shut-in and reduce the influence of afterflow and segregation.

Fair (1981) was the first to propose an analytical model that captures phase-redistribution effects in an exponential form. In this approach, phase-redistribution is treated as a time-dependent wellbore storage term in the diffusivity equation for an infinite, homogeneous, radial reservoir model. Key assumptions include a constant gas/oil ratio in the wellbore, negligible temperature effects, ideal and incompressible behavior for both liquid and gas, and a linear increase in pressure with depth. This exponential approach showed that even subtle phase-redistribution can distort early-time data. Later, Thompson et al. (1986) further advanced these interpretation techniques by categorizing the severity of phase-redistribution into three types, ranging from substantial to negligible redistribution effects (Type I, II, and III). Since the exponential model fails to capture the phase segregation effect when the pressure buildup shows a decreasing wellbore storage coefficient, Hegeman et al. (1993) explored changing wellbore storage scenarios (increasing or decreasing pressure); to extend Fair's original model. In addition, Y. Li et al. (2014) incorporated a cosine function to capture oscillatory wellbore storage, and Gholamzadeh et al. (2020) introduced a new physical model for capturing non-ideal wellbore storage.

Despite these advancements, accurately correcting for phase segregation in high water-cut or other challenging wellbore conditions remains a challenge. Many existing approaches require case-specific tuning or extensive parameter fitting, which can be time-consuming and may not eliminate early-time distortions. We address these challenges by presenting a novel, data-driven approach that leverages supervised, physics-guided convolutional neural networks (CNNs) to remove phase segregation distortions in vertical wells. We detail our methodology for generating training data through high-fidelity reservoir and wellbore simulations, describe the network architecture and training procedures, and demonstrate the effectiveness of the proposed solution on synthetic examples. Overall, our work aims to offer a more flexible and efficient means of mitigating phase-redistribution effects, to thereby enhance the reliability of PTA interpretations.

Methodology

Conventionally, obtaining reservoir parameters from a buildup test distorted by phase-redistribution begins with identifying the interval dominated by wellbore segregation via pressure derivative plots (Mattar & Zaoral, 1992; Olarewaju, 1990; Olarewaju & Lee, 1989a; Qasem et al., 2001, 2002; Thompson et al., 1986). Next, analysts match semi-log or log-log type curves or employ advanced history matching workflows based on analytical solutions of the diffusivity equation with two-phase fluid flow corrections (Fair, 1981; Gholamzadeh et al., 2020; Hegeman et al., 1993; Y. Li et al., 2014) to extract permeability, skin factor, and wellbore storage. However, these methods typically discard early-time data affected by multiphase wellbore effects and rely on simplistic corrections to capture complex fluid redistribution.

This work introduces a learning-based correction step that preserves the full pressure buildup record, and includes early time intervals dominated by fluid segregation (Fig. 1 outlines the main components of this workflow). First, we generate a comprehensive training dataset via high-fidelity simulations to produce paired pressure curves, with one set affected by phase-redistribution and the corresponding clean signals. We then train a neural network to take the distorted pressure signal and recover its clean counterpart. Once trained, this network removes the effects of phase-redistribution which allows conventional type curves and history-matching workflows to leverage the full dataset. By coupling data-driven preprocessing with established matching techniques, this approach retains the critical early-time information, enhances match quality across all time periods, and reduces the reliance on idealized two-phase storage models that try to incorporate phase-redistribution changes.

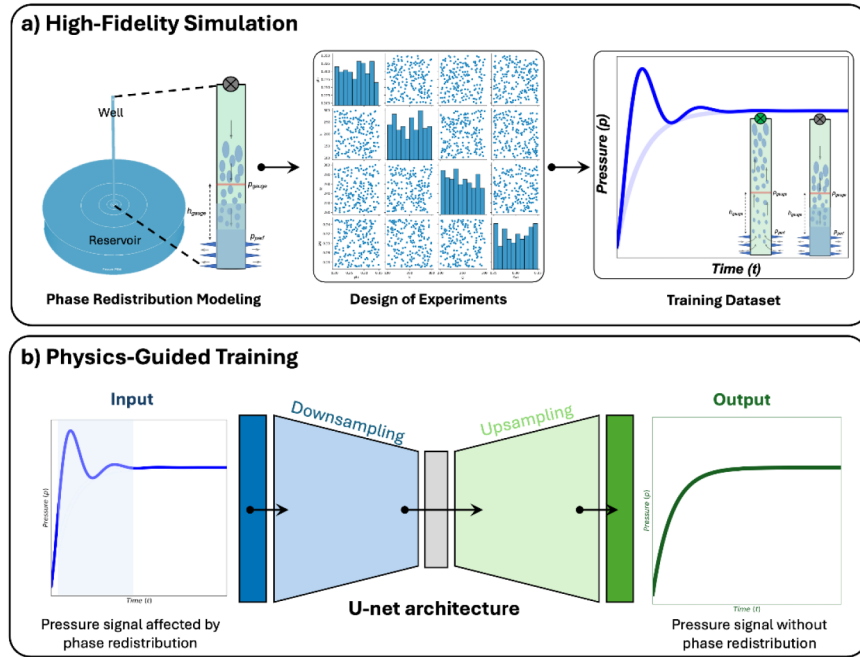


Figure 1—Workflow of the proposed methodology, which is comprised of two main stages. a) Stage one represents high fidelity simulation and dataset generation. Phase segregation is modeled on a radial grid coupled with fluid flow in the pipe using ECLIPSE. A design of experiments over key variables produces paired pressure signals, one affected by phase-redistribution and the corresponding clean baseline. b) Stage two represents the supervised training pipeline. A U-Net model takes as input the affected pressure, along with a user defined mask indicating segregation regions, and learns to predict the underlying clean pressure signal.

High Fidelity Simulation

Modeling phase-redistribution effects within the wellbore is both challenging and complex (S.-J. Li et al., 2023). Furthermore, coupling those effects with reservoir dynamics results in time-consuming simulations prone to convergence errors. Instead of a fully coupled two-phase reservoir-wellbore model, we employ a more straightforward, yet physically consistent, approach. We run a standard black oil simulation in ECLIPSE and represent the production well as a series of axial segments (Holmes et al., 1998).

Forward Modeling

For fluid systems, which typically consist of N_{ph} fluid phases and N_c components, and when $N_c = N_{ph}$ each phase consists of a single component, the immiscible multiphase flow system is governed by one equation per phase and can be described as follows:

$$\frac{\partial}{\partial t}(S_a \phi \rho_a) + \nabla \cdot (\rho_a \vec{v}_a) - q_a = 0. \quad (1)$$

Then taking into account the closure relationship, the fluxes $\vec{v}_a$ for each phase are given by a multiphase extension of Darcy's law:

$$\vec{v}_a = -\mathbf{K} \frac{k_{ra}}{\mu_a} (\nabla p_a - g \rho_a \nabla z) \quad (2)$$

Where ϕ is rock porosity, ρ_a the fluid mass density of the phase, S_a the saturation of the phase, and $\mathbf{K}$ the permeability of the system. Each segment captures the local bottom-hole pressure, emulating downhole sensors at multiple depths along the completion, including directly at the sandface. This configuration reproduces the interaction between reservoir flow and the measured pressure signature, and does not require explicit and complex wellbore modeling. This approach mirrors field practice where permanent gauges cannot be installed at perforations and provides a practical foundation for subsequent data-driven correction of phase-redistribution effects. The pressure equation for each segment (Holmes et al., 1998) considers the

pressure in the node P_n , the pressure in the adjacent node $P_{(n-1)}$, the hydrostatic pressure drop ΔP_H , the friction pressure drop ΔP_F , and the acceleration pressure drop ΔP_A :

$$P_n - P_{n-1} - \Delta P_H - \Delta P_F - \Delta P_A = 0 \quad (3)$$

In the present study, the reservoir is discretized as a single $10 \times 1 \times 1$ Cartesian grid in the radial, azimuthal, and vertical directions, and only considers water and oil phases. Radially, the model uses ten annular cells with inner radii starting at 0.40 ft and block thicknesses of approximately [1.75, 2.32, 5.01, 10.84, 23.39, 50.55, 109.21, 235.92, 509.68, 1101.08] ft, and a constant 10 ft thickness in the vertical. Each cell is assigned values for porosity ϕ , horizontal permeability $k_x(x)$, and vertical permeability k_z (with $k_k = 0.1 k_x$). The production well is represented by a multi-segment wellbore consisting of 41 equally spaced axial segments, each 5 ft long and 0.4 ft in diameter. The pressure drop is evaluated for each segment as the sum of frictional loss (based on mixture velocity and segment geometry), hydrostatic head change (from local fluid density and elevation difference), and a drift-flux correction for oil-water slip. Segment pressures are updated iteratively until the mass and momentum residuals converge. This tight coupling of detailed wellbore hydraulics with the reservoir's finite-difference solution ensures a more rigorous modeling of phase segregation, hydrostatic effects, and frictional losses throughout the system.

The key insight of this modeling approach is the ability to treat the pressure readings from different wellbore segments as if they were recordings from separate sensors, each distorted by phase segregation. Since these segments sit above the sandface, we correct their readings by accounting for the hydrostatic pressure drop. Hence, the equivalent sandface pressure affected by phase-redistribution P_{cs} is computed by taking the measured pressure at each segment or node P_n and adding the hydrostatic head, which is the product of the local fluid mixture density ρ_m , gravitational acceleration g , and the vertical distance from that segment to the sandface h_n . These adjusted values constitute our "distorted" pressure signals, while the direct simulation output at the sandface serves as the corresponding "clean" reference. Thus, we automatically generate paired input and target data for the subsequent learning-based correction.

$$P_{cs} = P_n + \rho_m g h_n \quad (4)$$

Design of Experiments

To assemble our training dataset, we first generate a representative set of 1000 simulations using a design of experiments based on Latin hypercube sampling (Hoteit et al., 2022; Shields & Zhang, 2016). Four reservoir and production variables are sampled uniformly to induce a wide range of phase-redistribution effects: porosity between 0.20 and 0.35, permeability between 100 and 300 mD, production rate between 200 and 300 barrels per day, and initial water saturation between 0.25 and 0.35. Each simulation produces pressure curves at multiple wellbore segments and the reference pressure at the sandface. To ensure a one-to-one correspondence between a single distorted signal and its clean counterpart, we chose the reading from segment fifteen as our distorted input and pair it with the sand face measurement as the target output. Segment fifteen lies midway along the completion and captures representative wellbore redistribution. This procedure yields 1000 input-target pairs for supervised training. Fig. 2 illustrates a selection of sample pairs from our dataset, showing the input pressure curves distorted by phase-redistribution, alongside their corresponding clean pressure targets.

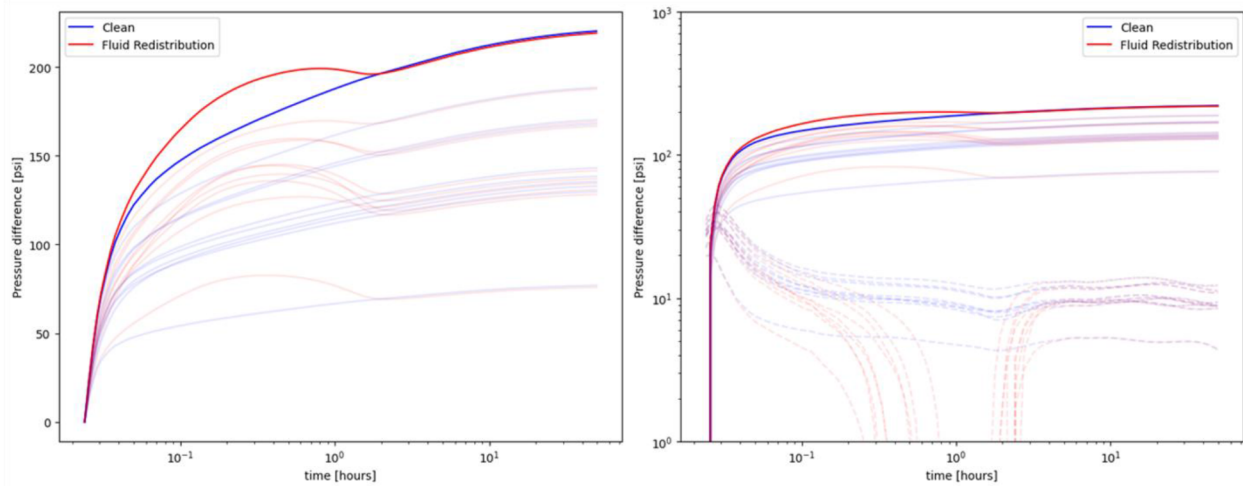


Figure 2—Subset of the training dataset showing pressure curves on linear (left) and log (right) scales. Red lines represent pressure signals distorted by phase-redistribution, and blue traces show the corresponding clean pressure targets. The right panel also plots the Bourdet derivative for each case.

Physics-Guided Training

We design a one-dimensional UNet to recover the clean pressure and Bourdet derivative curves from distorted pressure measurements (see Fig. 1b). The input of the network is composed of three channels: distorted pressure (affected by phase-redistribution and normalized between 0 and 1), the Bourdet derivative (normalized between -1 and 1), and a binary mask indicating phase segregation regions (composed of 0 and 1). The mask is generated by identifying intervals where the derivative falls below zero, which is an indicator of fluid segregation in the wellbore. During training, we add random noise to the distorted pressure and derivative. In the encoder path, each stage applies a convolutional layer with a kernel size of seven and the same padding, followed by batch normalization and a leaky ReLU activation, then a max pool by a factor of two. The bottleneck expands the feature depth to 256 channels before the decoder upsamples via linear interpolation applies a one-by-one convolution to halve the channel and concatenates with the corresponding encoder feature map. Two decoder stages mirror the encoder path, ending with batch normalization and a leaky ReLU. A final one-by-one convolution projects the two output channels, the clean pressure signal, and the clean Bourdet derivative. The pressure output is passed through a sigmoid and then rescaled to enforce a monotonic buildup curve between the initial and final measured values. The masking blends the network prediction and the original measurement outside segregation regions.

We use dropout with a probability of 0.2 after every batch normalization and activation throughout the model to regularize training. We optimize the network weights using the Adam optimizer with an initial learning rate of 0.01 and weight decay of $1e-5$. Our synthetic dataset comprises 1000 experiments, split into training, validation, and test subsets in a seventy-fifteen-fifteen ratio. The loss function combines the mean squared error and Huber loss on both outputs within the masked regions, and averages over the batch. Empirically, we find that similar performance can be achieved with fewer samples, which indicates that the model generalizes well even with a reduced training set. In compact form, we denote our U-Net model by f_{θ} . Given the distorted pressure signal p_{pr} , its Bourdet derivative b_{pr} , and the user-defined mask m , the network predicts the clean pressure p and the clean Bourdet derivative b .

$$(p, b) = f_{\theta}(p_{pr}, b_{pr}, m) \quad (5)$$

The combined mean squared error and Huber loss on both pressure and derivative outputs yields smooth convergence during training. Training and validation losses decrease steadily and remain closely aligned throughout the epochs. This behavior suggests a stable generalization and no signs of overfitting (see Fig. 3).

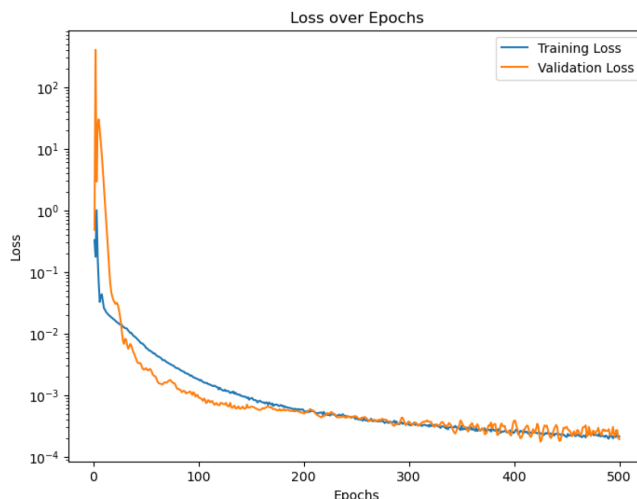


Figure 3—Training and validation losses evolution for the U-Net model. The loss is computed as the sum of masked mean squared error and Huber loss on both the pressure and Bourdet derivative predictions.

Results

To evaluate the trained network's performance prior to field deployment, we apply the network to a blind test dataset drawn from our synthetic dataset. This assessment quantifies the network's accuracy in recovering both the clean pressure signal and the Bourdet derivative, with the mask serving to focus the prediction on intervals affected by phase-redistribution. For illustration, Fig. 4 presents the results for two randomly selected test samples. On the left, we plot the normalized input (distorted pressure), the network prediction, and the ground-truth pressure on the same linear scale. The right panel shows the corresponding pressure buildup curves on a logarithmic axis. For clarity, only the pressure profiles are displayed, though the Bourdet derivative predictions exhibit similarly close agreement with their true values.

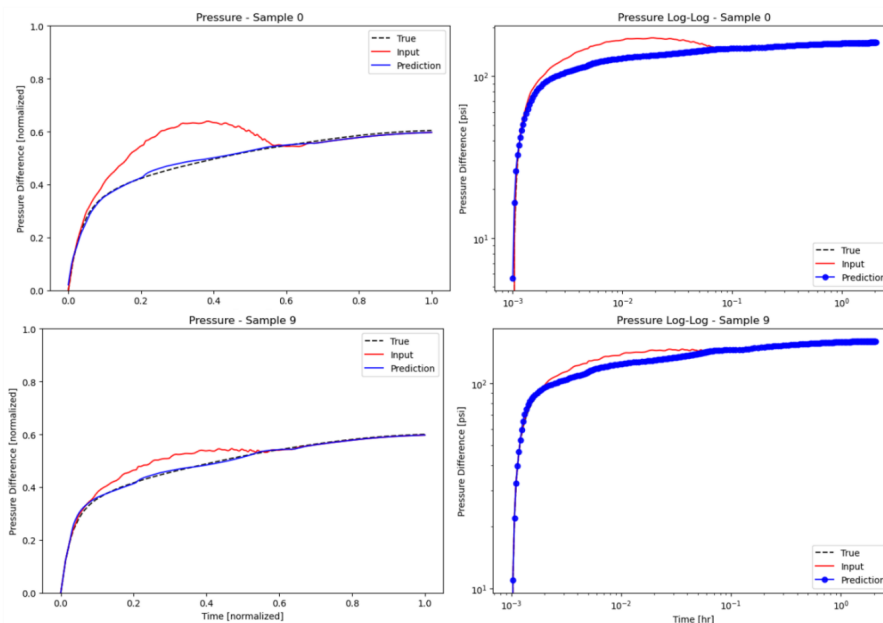


Figure 4—Blind test results for two randomly selected synthetic samples. Left panels show normalized linear plots of distorted input pressure, network prediction, and ground truth pressure. Right panels present the same buildup data on logarithmic axes. Although only the pressure profiles are shown, the network achieves comparable accuracy for the Bourdet derivative.

Discussion

This work shows that a neural network-based correction can recover the true sandface pressure from buildup test measurements affected by fluid redistribution in the wellbore. By combining a standard black oil simulation in ECLIPSE with a multi-segment well representation and a supervised U-Net model trained on synthetic experiments, our method preserves information across the entire pressure trace, including the early intervals dominated by phase segregation. The network delivers accurate results on blind synthetic tests from a vertical well with moderate water-cut. Conventional workflows either discard early time data or embed an analytical model within type curve or history matching frameworks that tries to capture phase-redistribution. Such approaches can bias estimates of permeability, skin, and wellbore storage by ignoring the first minutes of a buildup test or oversimplifying the analytical solutions of the DE. Our approach fundamentally differs, as it removes phase-redistribution effects from the pressure signal before conventional PTA routines. Once the corrupted intervals are corrected, standard analytical type curves or automatic history matching can proceed without the need for an explicit wellbore phase-redistribution analytical solution.

The presented framework opens new opportunities for accurate characterization in Pressure Transient Analysis. Future work could extend the method to three-phase flow by incorporating gas into the simulation and the training data. Also, future research could widen the range of reservoir and fluid properties to improve generalization. Finally, coupling full computational fluid dynamics simulations of the wellbore with the reservoir finite difference model could provide a more physics-guided foundation for network training and inference.

Conclusions

We demonstrate that a neural network correction can predict the corrected sandface pressure from buildup tests distorted by fluid redistribution in the wellbore. This study integrates a standard black oil simulation in ECLIPSE with a multi-segment well representation and a supervised UNet trained on synthetic experiments. Unlike traditional workflows that discard early time data or rely on analytical corrections within type curve or history matching, our method removes the distortion up front and then standard matching techniques without an explicit wellbore model can be used for further characterization. The network can process any time window of interest, focus corrections only where needed, and be applied iteratively to reduce residual artifacts. While our current implementation covers vertical wells with oil and water and a limited range of reservoir properties, extending the training set to include gas and deviated geometries and more detailed fluid dynamics will broaden its applicability. Overall, the proposed approach offers a more practical path to enhance field PTA for wells affected by phase-redistribution.

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